MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS

NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 7 14/10/2021

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We will discuss several questions from Chapter 5 and Chapter 6 in Ross' book.

Tutorial Questions

(1) Two fair dice are rolled. Find
  (i) the joint probability mass function of X and Y, and
  (ii) the marginal probability mass function of Y, and
  (iii) the conditional probability mass function of Y given that $X = 4$
  when
  (a) X is the largest value obtained on any die and Y is the sum of the values;
  (b) X is the value on the first die and Y is the larger of the two values;
  (c) X is the smallest and Y is the largest value obtained on the dice.

(2) Consider independent trials, each of which results in outcome i, $i = 0, 1, \dots, k$, with probability p_i, $\sum_{i=0}^{k} p_i = 1$. Let N denote the number of trials needed to obtain an outcome that is not equal to 0, and let X be that outcome.
  (a) Find $P\{N = n\}$, $n \geq 1$.
  (b) Find $P\{X = j\}$, $j = 1, \dots, k$.
  (c) Show that $P\{N = n, X = j\} = P\{N = n\}P\{X = j\}$.
  (d) Is it intuitive to you that N is independent of X?
  (e) Is it intuitive to you that X is independent of N?

(3) Suppose that X and Y are independent geometric random variables with the same parameter p.
  (a) Without any computations, what do you think is the value of

$$P\{X = i | X + Y = n\}?$$

  Hint: Imagine that you continually flip a coin having probability p of coming up heads. If the second head occurs on the nth flip, what is the probability mass function of the time of the first head?
  (b) Verify your conjecture in part (a).

(4) The joint probability mass function of X and Y is given by

$$p(1,1) = \frac{1}{8}, \quad p(1,2) = \frac{1}{4}$$
$$p(2,1) = \frac{1}{8}, \quad p(2,2) = \frac{1}{2}$$

  (a) Compute the conditional mass function of X given $Y = i$, $i = 1, 2$.
  (b) Are X and Y independent?
  (c) Compute $P\{XY \leq 3\}$, $P\{X + Y > 2\}$, $P\{X/Y > 1\}$.

(5) The joint probability density function of X and Y is given by

$$f(x, y) = c(y^2 - x^2)e^{-y}, \quad -y \leq x \leq y, \ 0 < y < \infty$$

  (a) Find c.
  (b) Find the marginal densities of X and Y.
  (c) Find $\mathbb{E}[X]$.

(6) The joint density function of X and Y is

$$f(x,y) = \begin{cases} xy & 0 < x < 1, 0 < y < 2; \\ 0 & \text{otherwise,} \end{cases}$$

where c is a constant.

(a) Are X and Y independent?
(b) Find the density function of X.
(c) Find the density function of Y.
(d) Find $\mathbb{E}(Y)$.
(e) Find $P\{X+Y<1\}$.

Assignment Questions

(1) Two dice are rolled. Let X and Y denote, respectively, the largest and smallest values obtained. Compute the conditional mass function of Y given $X = i$, for $i = 1, 2, \ldots, 6$. Are X and Y independent? Why?

(2) Jay has two jobs to do, one after the other. Each attempt at job i takes one hour and is successful with probability p_i. If $p_1 = 0.3$ and $p_2 = 0.4$, what is the probability that it will take Jay more than 12 hours to be successful on both jobs?

(3) The joint probability density function of X and Y is given by

$$f(x,y) = \frac{6}{7}\left(x^2 + \frac{xy}{2}\right), \quad 0 < x < 1, 0 < y < 2.$$

(a) Compute the density function of X.
(b) Find $P\{X > Y\}$.
(c) Find $P\{Y > 1/2 | X < 1/2\}$.
(d) Find $\mathbb{E}(X)$.
(e) Find $\mathbb{E}(Y)$.

(4) If X_1 and X_2 are independent exponential random variables with respective parameters λ_1 and λ_2, find the distribution of $Z = X_1/X_2$. Also compute $P\{X_1 < X_2\}$.

(5) The joint density function of X and Y is

$$f(x,y) = \begin{cases} x/5 + cy & 0 < x < 1, 1 < y < 5; \\ 0 & \text{otherwise,} \end{cases}$$

where c is a constant.
(a) What is the value of c?
(b) Are X and Y independent?
(c) Find $P\{X + Y > 3\}$.